Young AI Leaders Linz Hub

GenAI for Good (IEEE & ITU)

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Mr. Jan Korytar

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Application Form

Instructions

The GenAI for Good Challenge asks that participants send their narrative applications for Phase One in the provided template below. Once submitted, the application will be pre-filtered using a compliance checker and reviewed by the judges to determine up to three finalists for each focus area that will proceed to the next round.

For the three focus areas—Climate, Health, and Agriculture—IEEE and ITU, in partnership with in-country UN agencies and national ministries, identified three concrete use cases aligned to nationwide challenges, each with clearly defined user personas and problem statements.

All necessary information for the development of your application can be found in the following links.

- *Agriculture: AI-Powered Agricultural Extension Chatbot*
- *Climate: AI-Powered Extreme Weather Advisor*
- *Health: AI-Chatbot for Prevention and Control of Non-Communicable Diseases*

Please carefully follow the requirements below and complete the form.

Note: Do not include any images, videos, or links to such media in your submission, unless indicated otherwise.

Submission Agreement*

By writing your initials in the following box, you, the applicant, are affirming that you have reviewed, accepted and agreed to the GenAI for Good Challenge 2025
Official Rules.

PV

A.) Administrative Information

Project/Team Name*

Young AI Leaders Linz Hub

Primary Applicant Name*

Peter Velosy

Primary Applicant Email*
peter@petervelosy.com

Primary Applicant IEEE OU affiliation (if applicable)
N/A

Primary Applicant IEEE Member Number (if applicable)
N/A

Primary Applicant IEEE Member Grade (if applicable)
Does the primary applicant identify as:
Male

Co-Applicant Name*
Jan Korytar

Co-Applicant Email*
jan.korytar.ml@gmail.com

Co-Applicant is affiliated with which IEEE OU (if applicable)
N/A

Co-Applicant IEEE Member Number (if applicable)
N/A

Co-Applicant IEEE Member Grade (if applicable)
N/A

Does the co-applicant identify as
Male

List the other members of the team, indicating which one(s) are IEEE members:*
Viktor Maximilian Loreth, not an IEEE member

Location(s) of team members:*
Please select all that apply

Austria

Czech Republic (Czechia)

Which of the following Use Cases does your proposal address?*
Health

B.) Team Background and Expertise (15%)

Motivation*
Tell us about your team and your motivation to participate in this challenge.

We are an international team of motivated young software and ML engineers holding degrees in AI from the Johannes Kepler University in Linz, Austria. As members of the Young AI Leaders Linz Hub, we are committed to increasing the positive societal impact of bringing ML based solutions to places where it has the potential to create equal chances for all. This desire is our primary source of motivation. A further important reason for joining is the opportunity to work closely together with IEEE, ITU and Gambian stakeholders towards delivering a UX suited to the behavioral patterns of the target audience.

We are confident in our ability to deliver a final product of the highest quality, based on our IT industry experience, our experience building AI models, working with diverse stakeholders, and delivering projects on time. Our team brings together members from different backgrounds, but all connect with AI, and we actively collaborate with researchers from Johannes Kepler University Linz, which is internationally recognized for its research in machine learning and healthcare applications.

About Us*
Why do you think you are well positioned to build a good solution for the selected use case? Include any relevant experience with AI-related projects, your experience or expertise in domain areas relevant to the use case you will be working on, or other skills that could contribute to the success of your proposed solution.

We combine genuine interest, industrial experience, and a strong academic background, making us capable of delivering an excellent project. Please refer to our CVs for a detailed overview of our work experience. Aside from the GENIE.AI framework and GovStack, we already have experience with all other relevant technologies.

1) Peter Kristof Velosy
Peter is an experienced software engineer and architect, who currently works as a Product Owner for software components of the national digital healthcare system of Germany, and operates a freelance IT consulting business based in Vienna (Basilical e.U.). He is pursuing a MSc in Artificial Intelligence at the Johannes Kepler University Linz and holds two further master's degrees in Social Anthropology (University of Pecs) and in Cognitive Neuroscience (Budapest University of Technology) and has worked in research projects in both areas. This experience lends the team a holistic perspective for evaluating the solution as it is being built with regards to its societal acceptance and with regards to the psychology of habit formation.

2) Viktor Maximilian Loreth
Viktor's career path combines academic experience in AI with practical software engineering and real project delivery. He has built and deployed machine-learning systems, worked with agentic frameworks, and handled full-stack development for products used by thousands. His background shows that he is capable of delivering high-quality projects used by large user bases. He also leads AI-focused communities (e.g. as one of the coordinators of Young AI Leaders Linz) and has run independent applied-AI projects.

3) Jan Korytar

Jan is a freelance web developer with a degree from one of Europe's earliest and most respected AI programs. He has worked directly with artificial intelligence in real-world projects and contributed to discussions at AI and healthcare conferences. His web development experience spans diverse clients and technologies, giving him a strong mix of technical knowledge and practical problem-solving skills. Additionally, he supports nature preservation NGOs by strengthening their web presence and assisting them with field activities.

Please upload CVs (curriculum vitae) for your team members.*

You may also provide links to your LinkedIn profiles, author pages, or similar.

cvs_combined.pdf

Team Roles*

For each team member, detail their assigned role and explain how tasks will be allocated based on these roles. Additionally, describe how this distribution will help you achieve your goals and manage resources in the overall process.

We all have the necessary knowledge and capability to deliver the project, and we plan to collaborate on all major components. However, a distinction can be made regarding primary areas of responsibility:

1) Peter Velosy
Responsible for project and requirements management, designing and developing the native mobile apps (iOS & Android), the WhatsApp bot backend and for operations related concerns (CI+CD, cloud architecture, security).

2) Jan Korytar
Responsible for the RAG Ingestion Pipeline, the ingestion and preprocessing of datasets and for the coordination of the human expert based validation workflow for derived documents. He will also be responsible for assembling the core LLM pipeline including the integration of the new embedding & reranker models, the design of the system prompt, and the testing of the LLM pipeline.

3) Viktor Maximilian Loreth
Reponsible for designing our custom Django-based headless services and their data architecture (persistence + API-s), for developing the GovStack integration and for setting them up as LLM-callable tools to the LLM pipeline of the GENIE:AI framework. He will also be responsible for coordinating information exchange with stakeholders from The Gambia.

C.) Solution Concept (15%)

About the Solution (10%)*

Please provide an overview of the solution you intend to develop, including the rationale and the target users. Additionally, address the feasibility of completing the project during the challenge, its sustainability for future refinement or adaptation, and any other aspects that make your project stand out as innovative or impactful.

1) General, App and accessibility
We will build a multi-channel digital NCD assistant that delivers evidence-based information and practical, personalized guidance for non-communicable diseases (hypertension, stroke, diabetes, cancer, and chronic respiratory conditions). The service is accessible where people already are, WhatsApp as the primary outreach channel, with native Android and iOS apps and a lightweight web interface providing the richer, app-only features. For the scope of the current project, speech-to-text (STT) and text-to-speech (TTS) support

will be provided in English, with STT/TTS implemented as reusable GENIE.AI modules and applied to channels where technically feasible.

2) Target Group
Primary users are the general population of The Gambia: people at risk of NCDs and those living with chronic conditions who lack reliable, timely health information or easy access to professionals. The design prioritises low-bandwidth, mobile-first interactions and includes gender-sensitive content to address risk factors and care needs that affect women differently.

3) Rationale
Many Gambians do not have affordable, timely access to trustworthy NCD guidance. Health services are stretched and in-person counselling is limited. At the same time mobile phones and WhatsApp specifically are widely used. Delivering simple, evidence-based advice and behaviour-support nudges over these channels lowers barriers to care, helps people self-manage, and reduces demand on overloaded health systems.

4) Functional scope of the project
- Mobile-friendly, accessible, multi-channel chat interface: A chatbot experience optimized for low-bandwidth, mobile use on WhatsApp, with additional access via mobile app and web. The solution supports both text and voice input (English STT, where technically feasible) to address varying literacy levels.
- Vital Data and Habit Tracking, integrated with the mobile application (Data can be recorded using the chatbot, time series data collected can be exported (i.e. to provide a blood pressure or tobacco consumption graph to a doctor or a nurse).
- Patient risk assessment and risk profile tracking: A questionnaire, integrated into the chatbot, will assess the user's risk for the supported NCD-s and persist this information, providing it as a context to the chatbot, so that future chat interactions are aware of the patient's risk profile.
- Push notifications (via mobile app or via WhatsApp (as server-initiated messages)): Notifications include delivering the patient scheduled information related to their disease risk group (e.g. messages from or inspired by the WHO BHBM message libraries). In addition, government officials will be able to set up event notifications (e.g., reminders for screening days, prevention campaigns, or public health announcements).

5) Feasibility
The project is technically feasible within the timeframe of the challenge. By leveraging cloud platforms such as AWS, we can rapidly develop and deploy an MVP that covers the core channels and features described above. Existing building blocks allow us to focus on the NCD-specific logic and user experience instead of low-level infrastructure. The software engineering experience of our team enables us to use state-of-the-art AI coding agents in a way that results in highly readable code and a consistent system architecture (by providing detailed, source code level references to the coding models in the prompts, by extending prompts with a functional specification document, and by refining AI generated code by hand).

From a sustainability and scalability perspective, the solution is designed to be modular, maintainable, and easy to extend. Using a modern cloud-based tech stack (e.g., AWS managed services) allows the system to scale with demand while controlling operational costs. The content layer is separated from the technical implementation so that the advice of medical experts can be used to update guidance, add new NCD topics, or adjust messaging for new campaigns without major redevelopment. New advances in AI models, as well as new research and guidelines on NCDs, can be integrated over time by updating specific components rather than rebuilding the system from scratch.

User Journey (5%)*

Please provide example(s) of a user journey for the solution you intend to build. Refer to the personas described in your focus areas and respective use case to find expected user descriptions. Think from the user's perspective and highlight how the user will interact with the different functionalities of the solution, and how these interactions help resolve the user's pain points.

1) Bakary (citizen with NCD risk factors, WhatsApp)
Bakary, 42, is at his stall in Brikama and sees a poster with a WhatsApp number for the NCD Health Assistant. In the evening he sends "I have headaches". The chatbot replies in simple English and offers audio playback. It explains that the service is free and confidential, then asks a few short questions about his age, smoking, headaches, and family history.
Based on his answers, the assistant flags him as at risk for high blood pressure and tobacco-related disease. It explains in non-technical language how smoking and high blood pressure can cause headaches and dizziness and shows a simple low / medium / high risk indicator.
The chatbot offers two options: "Get tips to reduce or quit smoking" and "Learn how to check your blood pressure". Bakary chooses to reduce smoking first. Together they set a small goal (for example, two fewer cigarettes per day this week) and identify his main triggers at the market. At an agreed time of the day (e.g. after work), the chatbot sends short nudges on WhatsApp (e.g. with tips for stress reduction in ways other than smoking). When Bakary types "craving" the assistant responds with concrete coping tips.
At the end of the week, the assistant checks in, asks if he is ready to set a quit date, and shares nearby facilities and convenient times to measure his blood pressure. His pain points (no time for clinic visits, generic information, no personalised help) are reduced through WhatsApp chats that fit around his working day.

2) Fatou (NCD patient, mobile app + notifications)
Fatou, 52, has high blood pressure and is diabetic. At a clinic visit, a nurse helps her install the NCD Health App and inputs her disease risk values. The app interface is in English, but the chat interface supports STT/TTS and all the entire application is accessible, i.e. can be used with a screen reader, too.
During onboarding, Fatou inputs her conditions, mealtimes, and reminder times. The app then creates a simple routine: medication reminders, a short daily audio tip, and a weekly health check"notification. When a notification appears, she can tap "took medication" or "skipped" from the lock screen; if she skips, the app asks why and suggests practical strategies.
For diet questions, Fatou opens the app, types in the short description of a dish. The assistant answers her with a recommendation, how often she should consume it, and suggestions to make it healthier. She can also enter blood pressure or blood sugar values from the clinic; the app colour-codes them and explains what they mean in very simple language and audio.

3) Awa (Ministry of Health admin, campaigns and insights)
Awa, 36, a health promotion officer, has an option of obtaining an anonymized data set of chat transcripts along with demographics. (A future version can offer LLM based aggregates of this data).
She notices many questions about headaches and dizziness from one region and low reported blood pressure checks. She plans a three-week screening campaign and sends out a message with an invitation for screening. During the campaign she tracks how many users open, respond, and report having a reading taken, then uses these insights to refine future campaigns and target limited outreach resources more effectively.

D.) Solution Design (40%)

Extension to the GENIE.AI Framework (20%):*

As part of the challenge, participants will be required to use the **GENIE.AI framework** as the foundation for assembling their solutions. All code must be published to a dedicated project branch in the GENIE.AI Gitlab repository and made available under Apache 2.0, MIT, or another equivalent open-source license.

Solutions will be evaluated based on how effectively they extend and enhance the GENIE.AI framework. Participants are encouraged to contribute new features, improvements, or integrations that expand the framework's capabilities.

Examples of desirable contributions include (but are not limited to):

- Mobile application development;
- Speech synthesis and speech-to-text (SST-TTS) functionality;
- Integration with external databases and structured data workflows;
- Embeddable web components for the chatbot;
- Implement a user interface with popular social messaging platforms (e.g., WhatsApp, Facebook Messenger, Telegram).

Innovative features or extensions beyond these examples (particularly that are relevant to your respective use case) are also welcome and will be considered as part of the evaluation.

1) Mobile Application Development
The chatbot interface will run natively on both Android and iOS devices. The apps will provide optimized performance, offline capabilities where feasible, and user control over device-level features such as notifications. The chosen architecture (see below) allows for a later integration of on-device LLM-s and for a straightforward porting process to other device classes (e.g. wearables like smart watches or glasses).

2) Speech-to-Text (STT) and Text-to-Speech (TTS) Modules
The system will integrate STT/TTS functionality in English, tailored to handle the English variant commonly spoken in The Gambia. These modules operate across all communication channels and are developed as reusable GENIE.AI extensions. There is currently no publicly available STT model for Mandinka, and training one is outside the scope of this project and its funding.

3) Primary Third-Party Messenger Integration: WhatsApp
Since WhatsApp is widely used in The Gambia and does not require new app installations or web-based access, it serves as the primary interface. Users can interact with the system via text messages or voice notes. Not every feature, such as push notifications, can be provided through WhatsApp, and we expect the mobile applications to include all of these functionalities.

4) Time Series Data Tracker: A headless backend service, used in our case for tracking vital data and habits over time, driven by user interactions with the chatbot. This component will be designed in a way that makes it reusable for any other kind of time series data in future GENIE:AI applications (e.g. crop yield over time, temperature, etc.)

5) User Feedback Module: A GENIE:AI extension in which users can rate the quality of the chatbot's responses according to a configurable set of quantitative and qualitative criteria. These ratings are saved into a database along with the relevant (anonymized) chat transcripts, so that they can be used for RLHF (Reinforcement learning from human feedback) in a future iteration.

6) Fully Modular Architecture
Every component, mobile applications, messaging integrations, voice modules are implemented in a modular manner, ensuring clear separation of logic, maintainability, and compatibility with the GENIE.AI framework. This ensures that all of them, including the mobile applications, are easily reusable for GENIE:AI applications targeted at different use cases.

High-Level Architecture (20%):*

Please describe the **high-level architecture** of your project, demonstrating how you will leverage the GENIE.AI framework to assemble a functional prototype for your selected use case.

Your description should include:

- The main components of your system and their respective roles;

- The overall workflow and data flow;
- Tools, libraries, and services you plan to use;
- GENIE.AI components you will reuse;
- Any improvements or extensions you plan to implement;
- Additional components or integrations introduced;

The description should share architectural choices that make your solution innovative, efficient, and well-aligned with the selected use case. Your response should demonstrate a clear understanding of how the technical design supports the goals of your solution and contributes to the broader GENIE.AI ecosystem.

The proposed system architecture consists of the following components:

1) RAG Ingestion Pipeline
Beyond the data provided, we will ingest data related to further NCD-s, with an additional focus on mental health and a healthy lifestyle, as these are crucial factors for the prevention and the treatment NCD-s. Speficically, we will ingest the WHO Hypertension Treatment Guide, materials of the WHO HEARTS programme, the WHO Mental Health Gap Action Programme Guideline for Mental Health and Substance Use, the WHO Implementation Manual for Psychological Interventions, the ASAM Clinical Guidelines for Alcohol Use Disorder and the BHBM message libraries Doing What Matters in Times of Stress, mActive, mBreathe Freely, mCervical Cancer and mDiabetes and information documents about available healthcare services and service points in The Gambia.

These ingested primary documents will be complemented by LLM-generated and human expert-checked derived documents which transform texts geared towards medical professionals to texts geared towards patients. The usage of such derived documents will aid the chatbot beyond the system prompt in delivering information targeted to and understandable for patients. Specifically, we will generate a combined NCD and Mental Health risk assessment questionnaire, per-NCD patient FAQ-s, per-NCD fictive patient chats and per-NCD "if-then-rules" of treatment and prevention.

2) Chatbot
Our chatbot pipeline will replace the reranker and the embedding models of the GENIE.AI framework with newer models of the same families:https://huggingface.co/BAAI/bge-m3 and https://huggingface.co/BAAI/bge-reranker-v2-m3.

The chatbot's configuration and system prompt will ensure that the model takes on the role of a healthcare expert, that it cites its sources where applicable, that it has a mode in which it proactively guides through a risk assessment questionnaire, and that it allows for tool calling via MCP. Tools integrated will encompass GovStack compliant registration and appointment scheduling services (https://specs.govstack.global/) and a custom developed headless vital data and habit tracker service.

3) STT / TTS model
Our solution will include open-source STT and TTS models, which will be available in both the native mobile application and to an extend WhatsApp bot. Since English is an official language of The Gambia, and the Gambian variant of English has certain unique characteristics concerning its grammar and phonetics (see https://unipub.uni-graz.at/obvugrhs/content/titleinfo/4466136/full.pdf), model selection will be based on user acceptance testing with speakers of Gambian English.

4) Vital Data and Habit Tracking Service
The proposed solution encompasses a custom headless web service developed in Python, using the Django framework and a PostgreSQL database, capable of tracking the following data as a time series for each user: Blood pressure, tobacco consumption, mood, weight, habits defined using chatbot-provided instructions (e.g. regular meals, regular exercise, regular hydration, use of tobacco substitutes, etc.)

This service also tracks patient profile information which is derived from chat interactions and provided to future chats as context. Furthermore, this information controls the content of scheduled mobile push notifications / server-intiiated WhatsApp messages. The following information will be stored: Gender, date of birth, body height, region, risk status per condition / focus area (no risk / risk / problem already present: Hypertension and other CVD, smoking, diabetes, cancer, pulmonary diseases, mental health problems, physical activity, unhealthy diet, substace use (excl. tobacco))

5) User Feedback Module Backend: A web service written in Python (using Django) that collects the information provided by chatbot users in our custom chat user feedback extension (accessible via the web based chat interface, the WhatsApp bot and from the mobile app)

6) WhatsApp Bot Backend
Our WhatsApp bot will be implemented by another backend service, written in Python, using the Django framework and a PostgreSQL database. This service will generate its replies using our LLM pipeline and will deliver them using the WhatsApp Cloud API (https://developers.facebook.com/docs/whatsapp/cloud-api/reference).

7) Mobile Application
The supplied mobile application will be developed using native frameworks (SwiftUI for iOS, Jetpack Compose for Android). This decision has been made in order to offer the maximum degree of flexibility for integrating on-device language models (e.g. Apple FoundationModels, Google Gemini Nano) or supporting new device types (smart watches, smart glasses, etc.) in the future. The amount of additional effort arising from not using cross-platform frameworks (e.g. Flutter, React Native, etc.) is balanced by our choice of declarative and cross-device native UI frameworks. Their usage allows us to implement very similar architectures on both platforms, and results in code that is highly portable across device classes within the same platform (including smart glasses, smart watches, etc.)

Functions of the mobile application: Login and Registration (incl. SSO), Chat (incl. voice messaging and read-aloud), Chat templates (Anamnesis, Condition specific chat starters, Chat starters for GovStack services), Scheduled push notifications (Notifications are selected and derived from predefined message libraries according to risks and conditions stored by the Vitals and Habit Tracker service. The app can pre-generate / pre-fetch a number of notifications in its cache in order to deliver them even when offline), Tracking of vitals, habits and mood, Data and chart export for medical checkups, Integration with Apple Health and Google Fit, Rewards and Streak, Collect rewards and display streak information per focus area / risk class, Settings (Set push notification schedule).

8) Testing and Evaluation
The chatbot will be evaluated using a LLM-as-a-judge approach involving an array of LLMs different from the Granite model used. The judge model will evaluate answers for each test case based on a detailed, structured requirements specification of the relevant modules.

E.) Implementation and Risk (30%)

Impact (5%)*

What types of impact do you anticipate your solution could generate, and how would you practically measure them during an implementation? Impact Metrics for this Challenge should consider the relevant Sustainable Development Goals (SDGs) and their associated indicators by specifying at least one SDG specific indicator relevant to your selected use case.

Please propose methods and metrics for measuring the impact of your prototype on the country where it will be implemented. Unlike individual product performance metrics such as accuracy or recall rates, these measurements should focus on the broader impact to the society or the intended user community.

(17) SDGs are at the heart of the United Nations' 2030 Agenda for Sustainable Development. Each focus area in the GenAI for Good Challenge has an associated SDG as follows:

1. Health (SDG 3- Good Health and Wellbeing)
2. Agriculture (SDG 2- Zero Hunger)
3. Climate (SDG 13-Climate Action)

All SDG Targets include a list of "indicators," which are specific, observable, and measurable characteristics. These targets can be used to show changes or progress made towards the overall target. *Click here to view the full list of SDG Targets* and select the indicators you would like to base your impact metrics on.

Measurements can include qualitative results (e.g., questionnaires, interviews) and / or be supported by quantitative data (e.g., yearly crop yields). Ensure your measurements specifically target, but are not limited to, the impact outlined in Section C1 (About the Solution). All measurements must be clear, data-driven, and practically traceable

We primarily target SDG 3 – Good Health and Well-Being.

Target 3.4: Reduce premature mortality from non-communicable diseases by one-third through prevention and treatment.
We aim to track improvements using official statistics on mortality rates attributed to cardiovascular disease, cancer, diabetes, chronic respiratory diseases, and suicide. These indicators can be compared across several years following the deployment of the model.

Target 3.8: Achieve universal health coverage and access to quality healthcare services.
Our chatbot is not a complete solution to this global challenge, but it represents an important step in the right direction. Measurement can be conducted using questionnaires administered to both users and non-users of the chatbot, allowing comparison of perceived access, support, and health-related behavior.

Targets 3.c and 3.d: Strengthen the health workforce and build national capacities for early warning and risk reduction.
Our chatbot supports and relieves overworked healthcare providers by automating routine guidance, delivering reliable information, and helping users manage their health proactively. This contributes to stronger health-system capacity and improved community-level resilience.

In general we want to reach at least these goals in the first 6-12 months:
- 5,000–15,000 total users across WhatsApp + mobile app.
- 1,000–3,000 monthly active users (MAU)

Collaboration (5%)*

How would you work with experts in your selected focus area and local country partners? Please describe the ways in which you would include these relationships and perspectives in the development of the solution.

We already have an agreement with experts in AI and healthcare from Johannes Kepler University Linz, who are available to provide guidance throughout the project. To ensure that our solution is as usable and culturally appropriate as possible for the local population, we plan to include testing and feedback from people in the region as well as other key stakeholders. This collaboration will help us refine the system, validate our assumptions, and ensure that the final product effectively addresses real needs. The project will be developed through an iterative design cycle to guarantee alignment with user and stakeholder expectations.

Technical Implementation Approach (10%)*

Please outline your plan for developing the prototype. You may also mention any methodologies or specific open-source tools and frameworks you plan to use to complement GENIE.AI to ensure efficient implementation.
We will develop our prototype with an agile, iterative approach, using the SCRUM methodology (our team members all have experience with it, Peter is a trained CSPO (current certification renewal pending)).

The prototype scope includes: (1) a WhatsApp chatbot providing RAG-backed NCD guidance; (2) native mobile applications (iOS SwiftUI, Android Jetpack Compose) supporting chat, voice input, and a simple vitals & habit tracker; and (3) lightweight backend services implementing vitals and habit tracking, STT/TTS and notification services.

1) Architecture & components
Our custom backend services will be developed in Python (Django) with PostgreSQL as the canonical store. Retrieval-augmented generation (RAG) will use ArangoDB, embeddings from the BGE family, the Granite 3.3 8b LLM with a citation generation LoRA adaptor (all from HuggingFace Hub), and GENIE:AI-s LangChain based orchestration pipeline to combine document retrieval, reranking, and LLM response generation. We will integrate open-source STT/TTS models (e.g., Whisper / whisper.cpp or Vosk for STT, Coqui TTS for synthesis) and evaluate model performance on Gambian English during early tests. WhatsApp integration will use the Meta Cloud API and the mobile apps will be native to maximize future on-device capabilities.

2) DevOps, CI/CD & hosting
Code will be containerized with Docker and delivered via GitHub Actions CI/CD pipelines to managed cloud services (AWS ECS/Fargate or equivalent) to minimize operational overhead. Observability is provided by Prometheus/Grafana and Sentry for error monitoring. Static content and backups will use S3-compatible storage and RDS for managed databases.

3) Security & governance
We will implement encryption-in-transit and at-rest, role-based access, consent-driven data collection, and clear retention/deletion policies. Clinical data flows and any shared datasets will adhere to Ministry of Health data governance requirements and be scoped to minimize PII.

4) Testing & validation
Testing comprises automated unit and integration suites, an LLM-as-a-judge evaluation bench for content correctness, clinician review for medical accuracy, and iterative user acceptance testing in The Gambia (including STT/TTS checks with native speakers). Key metrics will measure system performance (uptime, latency), STT accuracy (WER on Gambian English samples), retrieval precision, and pilot impact metrics (MAU, completion rate, knowledge gain).

5) Accessibility (WCAG & WCAG mobile)

Technical Risk Mitigation (5%)*

What are the main potential risks that you foresee, and how do you plan to mitigate them? Please focus specifically on the risks to the technical implementation of the prototype as well as potential performance issues. Describe your strategies for mitigating these risks, focusing on practical and proactive measures, as well as contingency planning.

1) Scalability and Performance Issues
Risk: High message volume on WhatsApp or other channels may cause latency, slow STT/TTS processing, or system bottlenecks.
Mitigation: We will deploy the system on AWS using autoscaling groups, serverless components (AWS Lambda), and load balancing (ALB). This allows the system to scale automatically during peak demand. STT/TTS workloads are offloaded to GPU-optimized instances.
Risk: Outage of system components
Mitigation: Monitoring solutions will be set up for all application components so that outages can be reacted to within a short timeframe. Furthermore, we will use Cloudflare to mitigate the risk of a DDoS attack.

2) Data Integrity and Security Issues
Risk: Data loss or data breach
Mitigation: Regular scheduled backups (incl. regular restoration tests) will ensure data integrity even in case of a ransomware attack. An automated vulnerability assessment will be conducted (to the extent allowed by the infrastructure providers) to ensure the system's security. When the system goes into production and additional financial resources are available, beyond the current financial scope, a security operations team and the relevant policies must be created and automated vulnerability assessments must be extended through regular, complex penetration tests including social engineering approaches.

File Attachment Summary

Applicant File Uploads

- cvs_combined.pdf

Viktor Maximilian Loreth, BSc, PMP

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Education

Johannes Kepler Universität Linz — *Linz, Austria*
M.Sc. in Business Informatics (GPA: 1,8) — 03/2024 – ong.
- *Major: Artificial Intelligence in Business*

Johannes Kepler Universität Linz — *Linz, Austria*
M.Sc. in Artificial Intelligence — 02/2024 – ong.
- *Relevant Modules: Probabilistic Models, Deep Learning, Theoretical Concepts of Machine Learning*

Johannes Kepler Universität Linz — *Linz, Austria*
B.Sc. in Artificial Intelligence — 09/2021 – 03/2024
- *Relevant Modules: Mathematics, Numerical Optimization, Supervised Learning, Unsupervised Learning*
- *Bachelor Thesis: Evaluation of Interpretability Methods in Image Recognition*

HTBLuVa Salzburg: Electronics and Technical Informatics — *Salzburg, Austria*
- *Major: Electronics and Technical Informatics*

Work Experience

Vertice Ltd. — *Linz, Austria*
Backend Engineer — 10/2025 – 11/2025
Contributed to the development of an agentic AI framework using LangGraph and AWS. Left during probation period as role scope diverged from my objective of client-facing and strategic AI work.

Educational Leave for one year — *Linz, Austria*
Pursued independent projects in applied AI and entrepreneurship. Built software products and managed an AWS Seed application. Conducted field research for a startup in West Africa, analyzing local market conditions and technology adoption.

reqPOOL GmbH. — *Linz, Austria*
28 hours IT-Consultant — 08/2022 – 08/2024
- *Implemented IT strategies & ERP procurement (RFI, RFP) for companies with revenues of €700M+*
- *Handled the pre-sales process from the client-side including requirement engineering and project management*
- *Served as agile project assistant for software development project with a €1.4M yearly budget*

Johannes Kepler Universität – Institute of Digital Business — *Linz, Austria*
20 hours - Individual full stack developer with the Laravel Framework 7.0 — 03/2022 – 06/2022
- *Project: Digitalization of Elections with more than 3.000 users*
- *Responsible for database design, front- and backend development, and Python add-ons*

Red Cross — *Salzburg, Austria*
Full stack Developer as alternative military service — 03/2020 – 08/2020
- *Developed a custom CRM to automate contract creation (Laravel)*
- *Held sole ownership for the project*

Volunteering Experiences

Local Committee President – AIESEC in Linz — *Linz, Austria*
- *Head of student organization, 6k€ Revenue Sales* — *09/2021 – 04/2023*

Regional Leader & co-Founder – Young AI Leaders Linz — *LINZ, AUSTRIA*
- *Established a hub (in collab. with ITU and the United Nations)* — *01/2025 – 1/2027*
- *Responsible for organizing hackathons, challenges & projects*
- *Public speaker for AI roadmap and AI literacy*

Skills & Interest

Languages:	German *(Native)*, English *(C1)*, Spanish *(A2)*
Technical:	Python, FastApi, Pytorch, Numpy, scikit-learn, PHP, Laravel, Linux, C++, MsOffice Suite, BPMN, Jira, Confluence, SQL, AWS,
Certificates:	Certified Professional for Requirement Engineering, Professional Scrum Master 1, TOGAF Enterprise Architecture Foundation 1, Project Management Professional PMP, Driving License B

Github link: https://viktorlor.github.io/
LinkedIn link: https://www.linkedin.com/in/viktor-loreth

Jan Korytar, BSc

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Education

BSc in Artificial Intelligence Oct. 2020 - Jan. 2025
Johannes Kepler University Linz
MSc in Artificial Intelligence Jan. 2025 - Expected Jul. 2026
Johannes Kepler University Linz
Awards: Merit Scholarship for excellent grades 3×, Public choice award: Game Development Course

Skills

Languages : English C1, Czech C2, German B2
Programming languages: Python, R, SQL
Tools: Git/GitHub, MS Office, Google Analytics, WordPress, Streamlit, Make, LaTeX
Libraries: PyTorch, NumPy, Matplotlib, Seaborn, Backtrader Pandas, Scikit-Learn, SciPy, Altair

Work experience

Freelance Web Developer & grant advisor | *nasliberec.cz, csopklenice.cz* Feb. 2020 - Present
Acquired several government grants (totaling 30 000 EUR) for website creation for small businesses and NGOs. Developed new websites, modernized outdated ones. Managed web analytics, advertising, and email marketing tools.

Tutor for Reinforcement Learning course | *Johannes Kepler University Linz* Oct. 2023 - Feb. 2024
Grading assignments, addressing students' questions, and providing assistance related to the course content.

Data intern | *IPUR.cz, Institute for Sustainable Development* Mar. 2021 - Jun. 2023
Analyzed website visitor data, go-to person for all things MS Excel, led successful donor outreach initiatives (darujme.cz).

Research assistant | *Technical University of Liberec* Nov. 2018 - Jul. 2020
Assisted in shock electrodialysis research, participated in scientific meetings, and created scientific posters for presentations.

Other Experience

Bachelor Thesis | *An Approach to the REFUGE Challenge With Limited Compute Resources* Jan. 2024- Jan. 2025
Developed a custom, small, U-net for the segmentation of the optic nerve head in retinal images, utilizing the REFUGE dataset. Obtained scores similar to the SOTA models in the original paper.

Member of the Brainary department, Deputy secretary | *Neuron-ai.at* Mar. 2023 - Jan. 2025
Organizing workshops, AI challenges, and paper reading groups focused on artificial intelligence topics for students and public.

AI in Healtcare | *Blended intensive program* Dec. 2024
Took part in a programme in Pavia, focused on the use of AI in healthcare, it's opportunities, risks, and ethical implications.

Head of Heimsprecher_innen, floor treasurer | *KHG - Franz Jägerstätter* Oct. 2022 - Oct. 2024
Organization of events, representation of dormitory residents, and management of floor finances. Responsible for the annual Sommerfest with 800 participants.

Péter Kristóf VELŐSY BA MA MSc

Senior IT Consultant
Current residence: Vienna

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Professional Experience with Reference Projects

Requirements Engineer, Technical Product Owner

Timeframe: Apr 2022 -
Employer: RISE (Research Industrial Systems Engineering) GmbH (Vienna, Austria)
Client: Gematik (Digital Healthcare Agency of Germany)

Requirements engineering of the following RISE products (all being central or edge components of the German Healthcare Telematics Infrastructure):

- Central IDP (Identity Provider)
- Federation Master
- KIM 1.5 (Encrypted communications platform of the telematics infrastructure)
- HSKVS (Management software for the 'RISE High-Speed-Connector' – a high-performance multi-tenant security hardware for accessing the telematics infrastructure, used by hospitals, insurance companies and IaaS providers)
- TiaaS (Telematics Infrastructure as a Service, from 2024 branded as RISE TI-Gateway)

Senior Lecturer

Timeframe: Sep 2021 – Feb 2022
Client: IBS International Business School, Budapest/Vienna/Remote

Teaching software development courses in IBS's 'Masters in IT for Business Data Analytics' programme (a joint degree programme with the University of Buckingham)

CTO, Lead Architect

Timeframe: Jan 2019 – Jun 2020
Employer: Enernova GmbH (Luzern, Switzerland)

Enernova GmbH is a Swiss company founded by senior managers of Siemens Building Technologies. As such, its main focus is developing software products (mainly in the area of business process management / automation) for Siemens Building Technologies. My roles included the organization of the company's new (partially remote) development team including the design of software development rules, guidelines and workflows. In addition, I carried out technical product design and development on the following products:

- CEDB: Commissioning and Engineering DataBase: A business application for managing the data and logistics processes required for the deployment and testing of Siemens Desigo based building automation solutions (CUBA, Spring, Spring Integration, Activiti, Vaadin, PostgreSQL).
- Simcaat and QuickCheckTool: Web based mobile tools (PWAs) for carrying out technical check-ups on buildings in order to provide building owners with a price quote for new energy optimization solutions based on Siemens automation products (Angular, PWA, Spring, ReST,

PostgreSQL, Jira)

- MET: A Jira based tool suite integrated with other Enernova and Siemens products for managing problem ticket workflows related to the deployment and testing of building automation solutions, including an offline web client (PWA) for Jira that replicates its configurable data entry forms and workflows, and lets engineers record tickets in environments where no stable network connection is present (i.e. in basements where many technical devices of crucial importance are located).

Freelance IT Professional (Architect / Consultant / Developer / Instructor)

Timeframe:	Aug 2015 – Current
Client:	Contractor for various Europe and US based clients (in the Engineering, Healthcare and Academic sectors)

Projects (selected examples):

Google Netherlands BV, Allianz Technology GmbH

Google Netherlands develops the 'G4' toolkit, originally produced by Cornerstone BV, which is a set of analysis and conversion tools for automatically transforming legacy mainframe applications (written in legacy languages, e.g. COBOL or PL/I) to state-of-the-art technologies. In this specific projekt, where I worked as a senior consultant, we adapted the G4 engine to perform the conversion of Allianz's core business services ('ABS') from IBM PL/I to a cloud-native Java Spring application, thereby further developing G4's conversion logic as well as its Java based emulation layer (that serves as a drop-in replacement for several PL/I API-s, data types and language features) according to the requirements posed by Allianz's code base.

Braining Hub (Budapest, Hungary):

Lead instructor and designer of courses on Java (J2SE, J2EE, Spring), Angular, Node.JS, Laravel, Software testing and development methodology. Course sessions include internal bootcamp groups as well as externally held courses aimed towards mid-level to senior IT professionals (clients include: T-Mobile Hungary, Bosch Hungary, Intrum Justitia Zrt). Some courses included additional consultancy services for refactoring existing products of the company based on the new technological stack being taught.

VetaTek LLC (San Diego, USA):

Architectural design and development of MetaBase, a metadata based business application development platform (NodeJS, Angular 5, MSSQL, Azure Cloud, Azure Application Insights, TFS Continuous Integration)

Institute of Ethnography of the Hungarian Academy of Sciences (Budapest, Hungary):

Participation in computer-aided music analysis and ethnomusicological archive digitization projects (ongoing work, about to be published).

WSCAD GmbH (Bergkirchen, Germany):

Development of a single sign on (SSO) solution based on WSO2 Identity Server extended by custom built plugins (J2EE)

Development of the Building AR mobile application for Android, which is capable of creating an editable floor plan of a building by recording its dimensions in an Augmented Reality environment

InnoTrain Kft (Budapest, Hungary):

Design and development of new version of the Online Trainer platform, which provides an online „virtual personal trainer" service by using computer generated training plans based on digitally measured and monitored health data. (Java EE, Spring, GraphQL, AspectJ, Maven, Jenkins, NodeJS, PostgreSQL, MongoDB, Android)

IDBS E-Workbook and EMC Documentum® integration layer

Timeframe: Sep 2015 – Nov 2015
Employer: DACHS Computing and Biosciences GmbH (Budapest office)
Client: Novartis AG, Basel, Switzerland

Architectural design and development of a synchronization middleware between IDBS E-workbook, where medical research data is generated, to EMC Documentum ®, where research documentation goes through the review and approval process. (Java EE, Spring, Oracle SQL DB, WebLogic)

VirtualEDI (EMC Documentum® Client)

Timeframe: Mar 2015 – Sep 2015
Employer: DACHS Computing and Biosciences GmbH (Budapest office)
Client: Novartis AG, Basel, Switzerland

A client application for an EMC® Documentum 2 system, developed specially for Novartis Pharma GmbH for worldwide use, which manages documentation and handles document lifecycles for medical research experiments.

EcoLane (fleet management and route planning)

Timeframe: Oct. 2014 – Mar 2015
Employer: ITWare Kft.
Client: EcoLane LLC, Espoo, Finland

EcoLane develops software which is mainly used by ambulance dispatchers in the United States. The present development involved re-writing their present one-tier PHP application to a three-tier Java EE system with a Ruby on Rails front-end.

MVNE (Mobile Virtual Network Enabler)-capable CRM development and GSM systems integration

Timeframe: Feb 2014 – Oct 2014
Employer: ITWare Kft., EuroMACC Kft.
Client: Ventocom GmbH, Vienna, Austria

Ventocom is a virtual mobile network enabler (MVNE) in Austria, founded by the past leaders of Orange Telecom. Our collaboration with them involved the development of an enterprise CRM system to allow the creation and management of any number of virtual mobile network operators (MVNO-s). I was in charge of designing and developing the integration with T-mobile's Home Location Register (HLR) and Mobile Network Portability (MNP) systems, as well as the subsystem for processing and generating SMS and USSD messages, in close cooperation with T-Mobile Austria's engineers. As per today, this system serves several virtual mobile operators including Allianz-SIM (https://www.allianz-sim.at/) and Hofer Telekom (https://www.hot.at/)

TransRound Android application translation framework

Timeframe: Oct 2013 – Mar 2015
Employer: ITWare Kft.
Client: TransRound Kft.

This is a framework which is capable of altering the UI of Android applications as to allow end users to translate and localize applications from within the application itself. The system is built by taking advantage of the AspectJ technology. I worked on the development of the core AspectJ component, the mobile application, and designed the backend and web services layer (collaborative translation facilities, machine translation integration, license management, etc.)

Mobile and backend development with Arteries Studio Kft.

Timeframe: Jan 2013 – Oct 2013
Employer: Arteries Studio Kft.

Cardy Android Application: Software which allows controlling Cardy ECG devices from mobile phones and tablets, displays measurement results, and synchronizes measurement data with a cloud back-end.

Fish and Chips E-Magazine Reader: This is an E-Magazine reader application for Android and iOS, where I was involved with developing the applications' backend.

ScooterMonitor.nl Android Application and Backend: This application enables users to track vehicles equipped with GPS, and it also provides users with diagnostic information acquired via the CAN bus protocol. I was one of the developers of the Android application and I designed its REST API protocol.

E-fit and Online Trainer (OT) Services: The first (proof-of-concept) version of the applications which later turned to InnoTrain's new product (see above).

Freelance projects (a selected few examples)

Timeframe: 2008-2012

FISEC International Student's Olympic Games: Online registration and result management system, where schools could register their competitors and teams, and referees could store their results, which were then populated as a searchable database.

Magnalex Kft: Interactive distance education portal based on Moodle. The Moodle base system has been extended with custom developed plug-ins, and with an enrollment management system, which reads and processes tuition fee payments from exported online banking account statements. These extensions also enable the management of company licenses, where users can enroll into courses if they access the site from within a given IP range. Students can download video based training material from the portal, which have their own installer, and which also require an online activation, protecting them against unauthorized use.

Oktatovideok.hu: Copy protection system using hardware identification and online activation to protect IT training materials published by this company.

Central Institute for Physics, Budapest (KFKI): Data processing system for ordering and evaluating electron cannon experiments.

Areas of Expertise

Courses and Certifications

Introduction to Accessibility and Inclusive Design (Coursera/University of Illinois at Urbana-Champaign)	2023
Design Thinking: Insights to Inspiration (Coursera/University of Virginia)	2023
Google SEO Fundamentals (Coursera/UC Davis)	2023
Certified Professional in Requirements Engineering - Foundation (IREB)	2023
Certififed Scrum Product Owner (Scrum Alliance)	2022
Neuromatch Academy - Computational Neuroscience	2021
Machine Learning (Coursera/Stanford, Andrew Ng)	2017
Embedded Systems (EDX/UTAustin, Valvano/Yerraballi)	2016
R Programming (Coursera/Johns Hopkins, Peng/Leek/Caffo)	2015
Software Security (Coursera/UMaryland, Michael Hicks)	2015

Key Soft Skills

An enthusiastic curiosity
An entrepreneurial mindset

The ability to inspire and to motivate others
A drive to think about interesting solutions to problems, human or technical
The openness to learn, teach and to listen

Programming Languages, Technologies and Frameworks

Java/Kotlin	Java SE Android development Spring Framework EJB CUBA Platform Struts
Swift	iOS development (SwiftUI + UIKit)
Ruby	Ruby on Rails
Python	Django PyTorch Scikit-Learn
Javascript/Typescript	Node.JS Angular React Vue.JS jQuery Next.js
PHP	Symfony Laravel phpUnit
C	
C++	
C#	.NET (Desktop+Web)
R	
Matlab/Octave	

Databases

PostgreSQL (with mid-level pl-pgSQL experience)
Oracle (with basic plSQL experience)
MySQL/MariaDB
SQLite
MongoDB
Neo4J
Redis
Firebase (Cloud FireStore)
CouchBase

Miscellaneous IT skills

Linux system administration
Knowledge of GSM specific protocols (especially SMPP)
Knowledge of the FIX (Financial Information Exchange) protocol
Issue trackers: Jira (with experience in plugin development), BitBucket, Confluence (administration + extension development) Redmine, Mantis, Bugzilla (administration)
UML
Experience with Agile methodologies
Version trackers: GIT, SVN, Mercurial, CVS
Testing tools: JMeter, SoapUI, LoadUI, Selenium, Cucumber
Build tools: Ant, Maven, Gradle, Webpack, Gulp
Machine learning (intermediate level, R, Python)
Containerisation (Docker, Kubernetes, OpenShift, Istio)
Cloud providers (AWS, Azure, GCP, IBM Cloud)

Education

Timeframe:	2025-
Institution:	Johannes Kepler University, Linz
Degree:	MSc in Artificial Intelligence

Timeframe:	2019-2023
Institution:	Budapest University of Technology
Degree:	MSc in Computational and Cognitive Neuroscience

Timeframe:	2017-2020
Institution:	University of Pécs
Degree:	MA in Cultural Anthropology

Timeframe:	2014-2020
Institution:	Vilmos Apor College
Degree:	BA in Choral Conducting and Church Music

Language Skills

Hungarian	Mother tongue
English	Advanced (C2, IELTS overall score: 8.5/9)

German	Advanced (C1, Goethe Institut + ÖSD)
Dutch	Beginner
French	Intermediate

Miscellaneous skills

Driving license	Category „B"